

SMIT

AI and Data Science Course

Batch 9

Final Project

AI Recruitment Assistant Dashboard for HR

Overview:

You will build a working web application that helps an HR team screen resumes automatically. The app takes one or more resume PDFs plus a job description, uses an AI model (Gemini) through LangChain to analyze each resume, and produces a summary, a skill match, a match score, interview questions, and HR feedback.

What To Build

- A Streamlit web app where HR uploads resumes and a job description.
- An AI pipeline that analyzes each resume against the job description.
- A ranking table that scores and sorts all candidates.

HR Dashboard Features:

HR Dashboard

- └─ Upload Job Description (PDF)

- └─ Upload Resume(s)

- └─ Single Resume

- └─ Multiple Resumes

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- └─ AI Analysis

- └─ Resume Summary

- └─ Match Score

- └─ Missing Skills

- └─ Interview Questions

- └─ HR Recommendation

- └─ Justification

- |

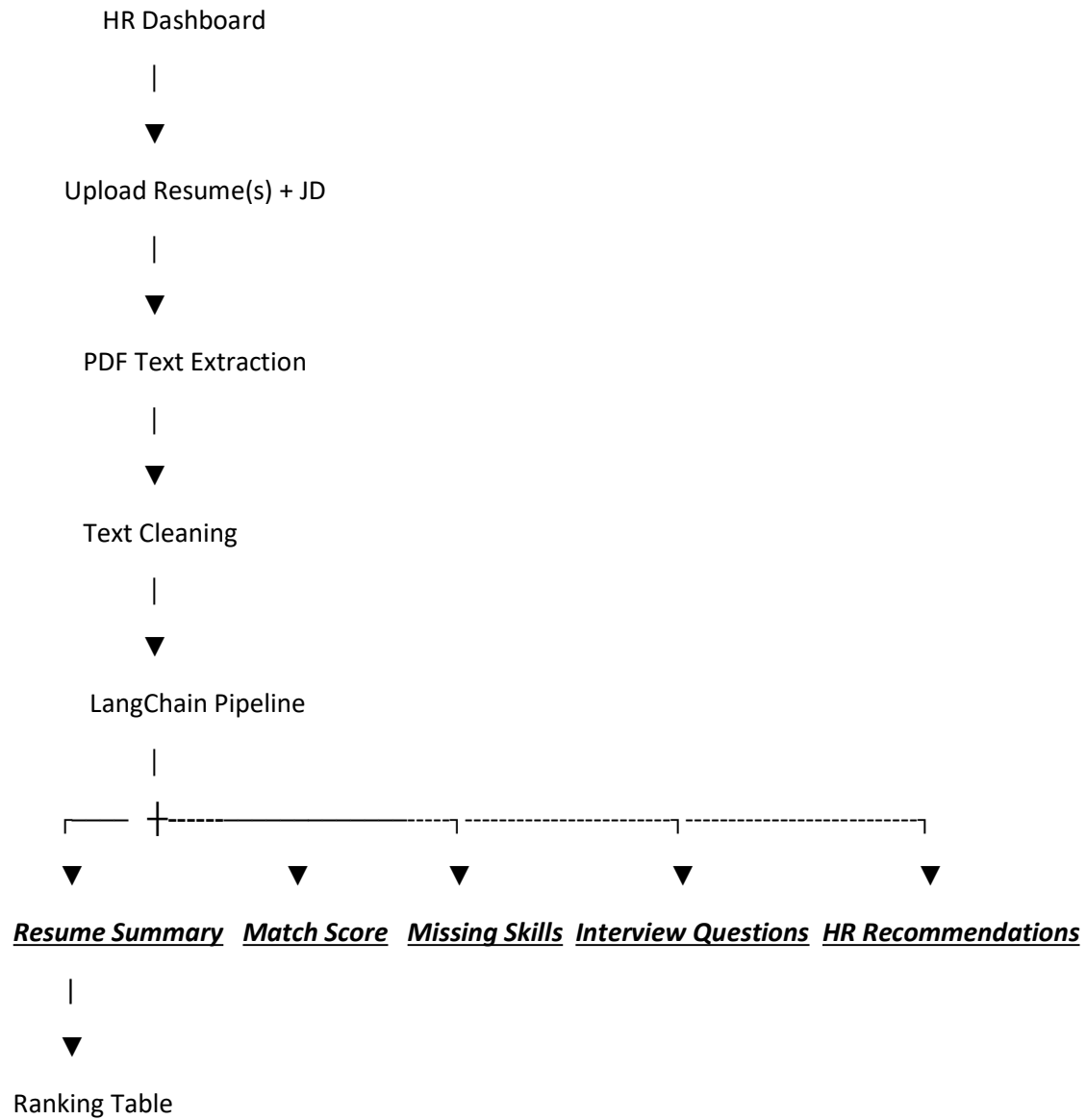
- └─ Ranking Dashboard

- └─ Candidate Comparison

- └─ Rank Candidates

- └─ Export CSV

System Architecture:



Module 1

Build the UI

Create the dashboard.

Sidebar

Upload Job Description

Upload Resume

Main Page

Analyze Resume Button

Only UI.

Module 2

Read PDFs

Workflow

Upload PDF

↓

Extract Text

↓

Display Text

Output

Resume Text

Module 3

LangChain

Pipeline

PromptTemplate

↓

LLM

↓

OutputParser

Module 4

Resume Summary – Chain 1

Goal: ask the LLM to pull out education, years of experience, and key skills.

Prompt: <write your own>.

Expected output

```
Candidate Summary
- BS Computer Science

- 3 years experience

- Python

- TensorFlow

- NLP

- Computer Vision

- Built AI Chatbots
```

Module 5

Match Score – Chain 2

Goal: compare resume skills against the job description and split them into Matching, Missing, and Extra.

Prompt: <write your own>.

Expected output:

```
Matching Skills: Python, SQL, TensorFlow
Missing: Docker, Azure
Extra: Power BI
```

Module 6

Match Score – Chain 3

Goal: produce a single percentage score in range (0-100%).

Prompt: <write your own>.

Output

Overall Match

89%

Module 7

HR Recommendation – Chain 4

Goal: give a hiring recommendation (i.e Hire or Rejected) and a short justification.

Prompt: <write your own>.

Output

Recommendation

Hire

Reason

Excellent ML knowledge

Good Projects

Strong Python Skills

Module 8

Interview Questions – Chain 5

Goal: generate technical and HR interview questions tailored to this resume and JD for the Hire candidates.

Prompt: <write your own>.

Expected output:

```
Technical: Explain Random Forest. What is TensorFlow? CNN vs RNN?  
HR: Tell me about yourself. Greatest achievement? ...
```

Module 9

Structured Output (JSON)

Instead of plain text, return JSON.

Example schema

```
{
  "summary": "...",
  "score": 88,
  "missing_skills": [],
  "interview_questions": [],
  "recommendation": "...",
  "justification": "..."
}
```

This makes the application more reliable and easier to display in Streamlit.

Module 10

Ranking Candidates

Objective: Process every uploaded resume and produce a single ranked comparison table.

Candidate	Score	summary	Missing Skills	Recommendation
Ali	95			Hire
Sara	90			Hire
Ahmed	85			Interview
Bilal	72			Reject

Module 11

Download Results

Export

CSV

Suggested Project Folder Structure

AI-Recruitment-Assistant/

```
|
|— app.py
|— requirements.txt
|— .env
```



```
├── components/
│   ├── sidebar.py
│   ├── uploader.py
│   └── ranking.py
├── utils/
│   ├── pdf_reader.py
│   ├── prompts.py
│   └── parser.py
├── ai/
│   ├── llm.py
│   └── chains.py
├── data/
└── outputs/
```

Final Deliverables

Before submitting, confirm you have every item below.

1. GitHub Repository

- Complete, working source code
- README.md explaining setup and usage
- requirements.txt
- Project screenshots

2. Streamlit Application

- Fully functional app

3. Sample Data

- At least 5 sample resumes
- At least 2 sample job descriptions

4. 10-Minute Demo Video (YouTube)

Your video must cover, in order:

- Project introduction
- Folder structure
- Code explanation
- LangChain + LLM workflow

- Prompt templates
 - Live demo: resume upload
 - Live demo: job description input
 - Live demo: resume summary
 - Live demo: skill matching
 - Live demo: candidate ranking
 - Live demo: HR feedback
 - Live demo: interview questions
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Future Implementations

For advanced consider these enhancements:

- **Radar Chart Comparison:** Visualize candidates across categories like Skills, Experience, Education, Projects, and Communication.
 - **Interactive Candidate Cards:** Expandable panels with detailed AI analysis for each applicant.
 - **Chat with Resume:** Allow HR to ask follow-up questions (e.g., "Does this candidate have leadership experience?") using Retrieval-Augmented Generation (RAG).
 - **Voice Summary:** Generate a short spoken summary using text-to-speech.
 - **Email Draft Generator:** Automatically draft interview invitations or rejection emails based on the recommendation.
 - **Authentication:** Add login functionality for HR users.
 - **Database Integration:** Store uploaded resumes and analysis history in SQLite or PostgreSQL.
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